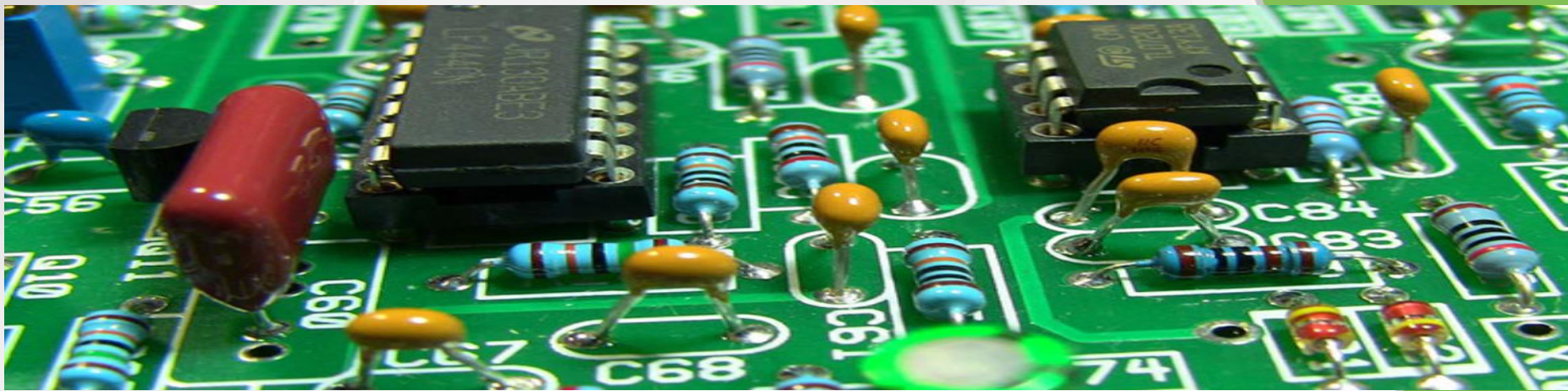




62732 Analog electronics



Technical University of Denmark

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62732 IoT

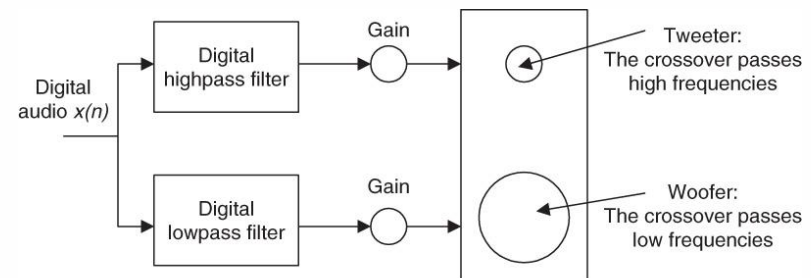
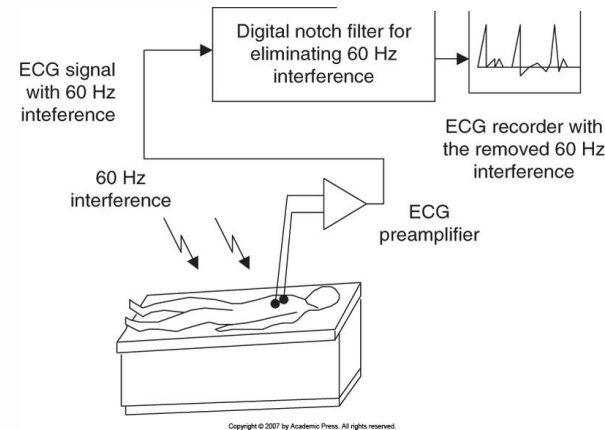
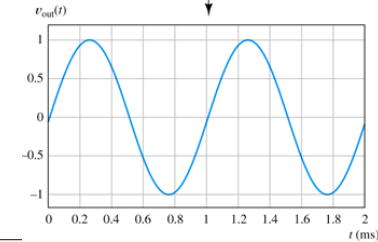
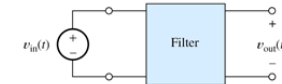
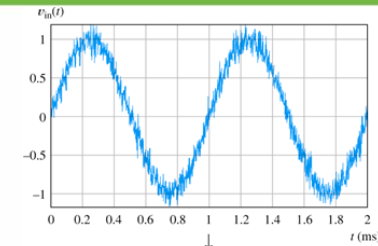
Slide 1

Filters

- Filters are two-port networks
- Filters allow certain frequencies to pass and reject others.
- E.g. an ECG amplifier allows approximately 0.05 to 100 Hz and notches 50 Hz (60Hz)

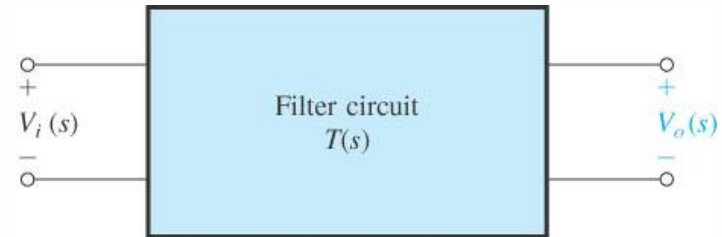
Filters are very important and they are used in many applications.
Some examples...

- Loudspeakers include filters in either HW or as digital filters.



Filters

- Linear filters can be represented by a general 2-ports network.
- Filter transfer function $T(s)$:
- The filter transmission set $s = j\omega$
- Expressed in terms of magnitude or attenuation in dB



$$T(s) \equiv \frac{V_o(s)}{V_i(s)}$$

$$T(j\omega) = |T(j\omega)|e^{j\phi(\omega)}$$

$$G(\omega) \equiv 20 \log |T(j\omega)|, \text{dB}$$

$$A(\omega) \equiv -20 \log |T(j\omega)|, \text{dB}$$

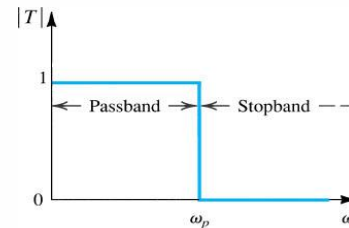
The filters can change the amplitude and phase of the input signal, but NOT the signal's frequency.

$$v_{\text{out}}(t) = 6 \cos(2000\pi t + 70^\circ)$$

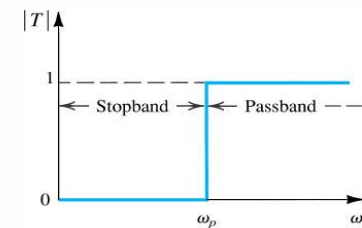
Filters

Real world filters cannot go from passband to stopband as step as show on these ideal prototype filters.

Filter types:



(a) Low-pass (LP)



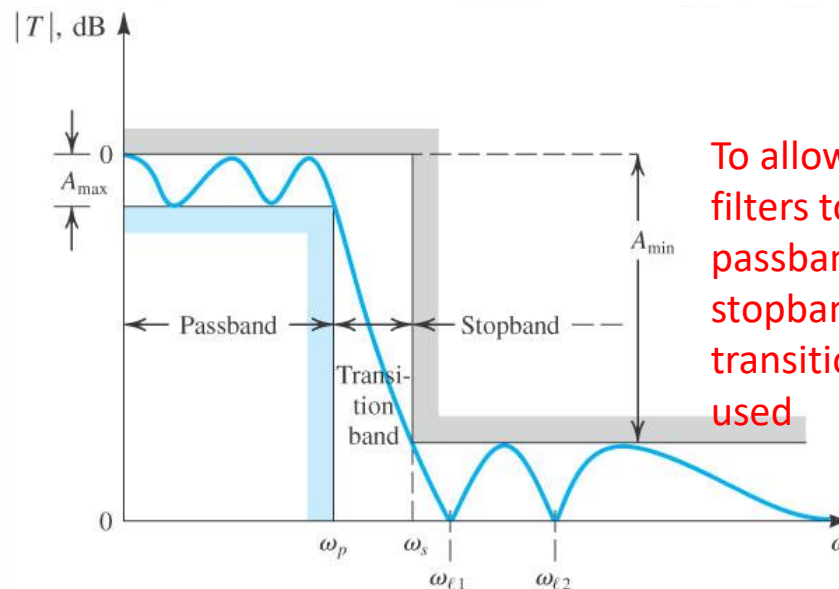
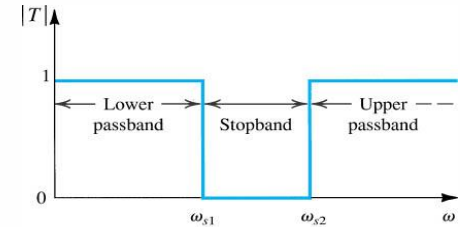
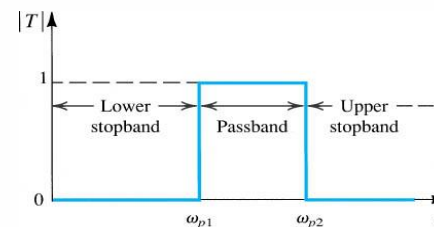
(b) High-pass (HP)

“Real world” filter specification:

- $A_{\max}$ typically ranges from 0.05 dB to 3 dB.
- $A_{\min}$ can range from 20 dB to 100 dB.

Requirements:

- Passband edge ω_p
- Maximum allowed variation in passband transmission $A_{\max}$
- Stopband edge ω_s
- Minimum required stopband attenuation $A_{\min}$

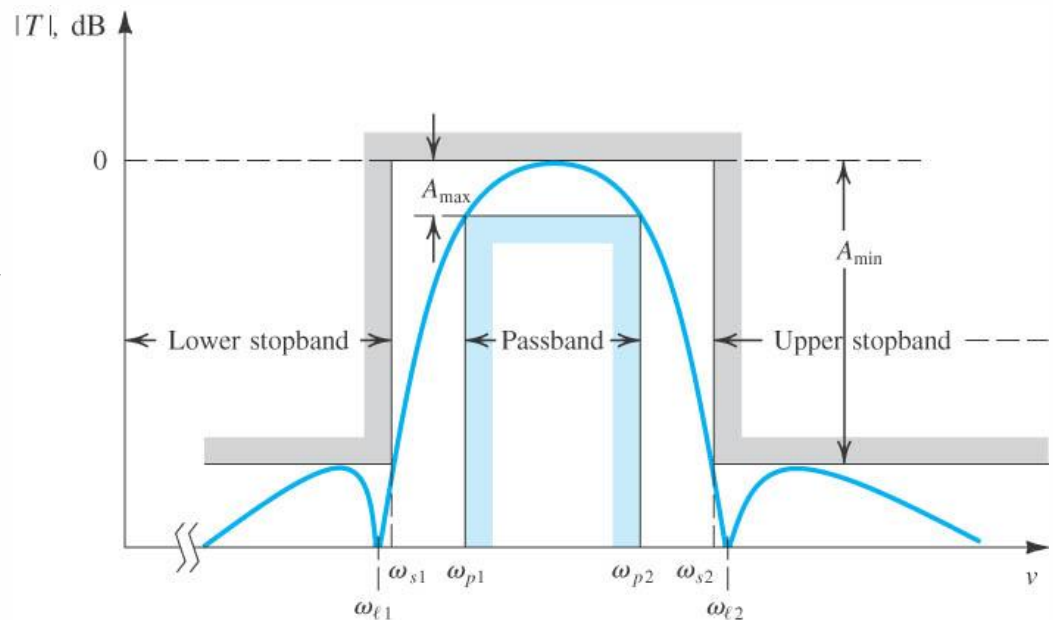


To allow real-world filters to go from passband to stopband a transition band is used

Filters

- The more tightly one specifies the filter requirements, the higher order and more complexity is required.
- The phase response of the filter is of interest in some applications, however, this is not in focus now.
- An example of a “realistic” filter design

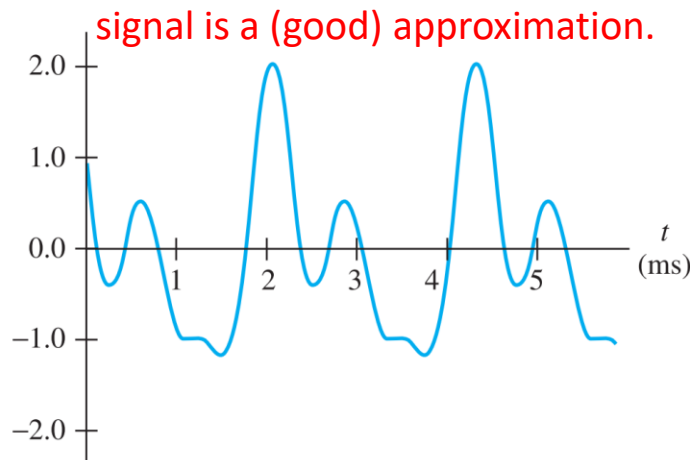
A rule of thumb says that the higher filter-order the more steep the filter-transitions can be.



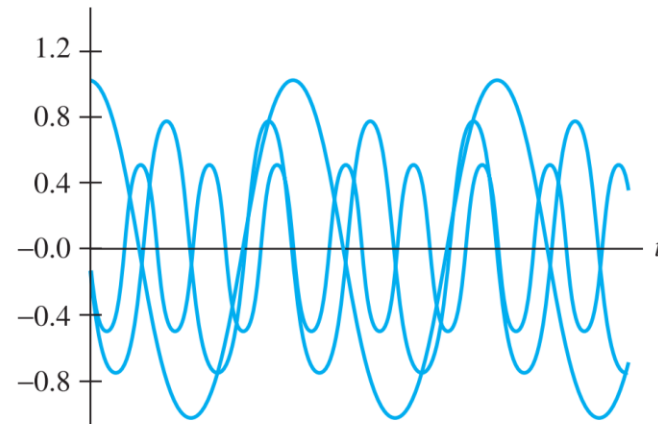
Intro

Fourier Series

Remember: most periodic signals can be split into a Fourier series. This means that it is possible to pass a periodic signal through a filter by passing each of the sinusoidal components through it and add them after the filter to get the filtered original signal back. Often it is not possible to pass all the frequencies through the filter, i.e. the filtered output signal is a (good) approximation.



(a) Music waveform



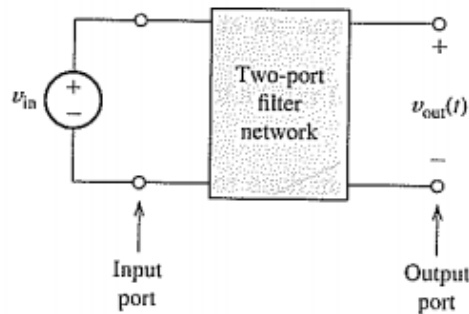
(b) Sinusoidal components

Figure 6.1 The short segment of a music waveform shown in (a) is the sum of the sinusoidal components shown in (b).

Filters

- The transfer function can be described with phasors:

$$H(f) = \frac{V_{out}}{V_{in}}$$



- Notice if more inputs at the same time superposition is used, i.e. handle each input on its own...

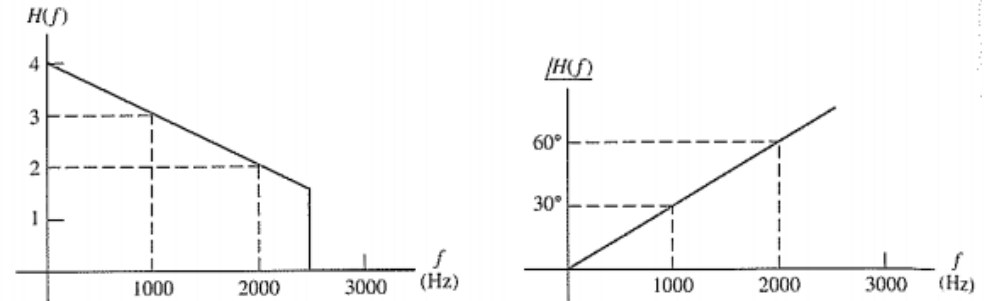


Figure 6.4 The transfer function of a filter. See Examples 6.1 and 6.2

$$v_{in}(t) = 2 \cos(2000\pi t + 40^\circ) = 2 \cos(2\pi 1000t + 40^\circ)$$

$$H(1000) = 3 \angle 30^\circ = \frac{V_{out}}{V_{in}}$$

The phasor for the input signal is $V_{in} = 2 \angle 40^\circ$, and we get

$$V_{out} = H(1000) \times V_{in} = 3 \angle 30^\circ \times 2 \angle 40^\circ = 6 \angle 70^\circ$$

Thus, the output signal is

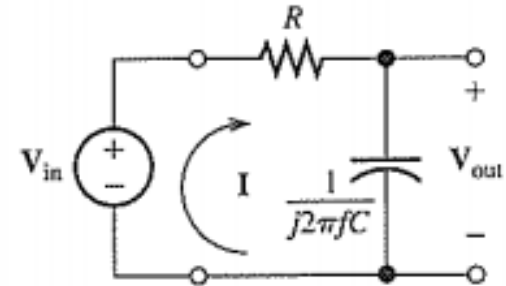
$$v_{out}(t) = 6 \cos(2000\pi t + 70^\circ)$$

1' order lowpass filters

- 1' order lowpass filter with a resistor and a capacitor.

- Let's find the transfer-function:

$$H(f) = \frac{V_{out}}{V_{in}}$$



- Using the voltage-divider equation:

$$V_o = V_i \frac{\frac{1}{Cs}}{\frac{1}{Cs} + R} = V_i \frac{1}{RCs + 1} = V_i \frac{1}{j\omega RC + 1} \text{ (where we only consider the } j\omega \text{ part of } s, \text{ i.e. } s = j\omega)$$

$$\text{Thus: } H(f) = \frac{V_o}{V_i} = \frac{1}{j\omega RC + 1} = \frac{1}{j2\pi f RC + 1} \text{ remember } \omega = 2\pi f$$

$$\text{But this can be written as: } H(f) = \frac{1}{j\left(\frac{f}{f_B}\right) + 1} \text{ if } f_B \text{ is defined as } f_B = \frac{1}{2\pi RC}$$

$$H(f) \text{ can be rewritten in polar form: } H(f) = \frac{1}{\sqrt{\left(\frac{f}{f_B}\right)^2 + 1}} e^{-\arctan\left(\frac{f}{f_B}\right)} = \frac{1}{\sqrt{\left(\frac{f}{f_B}\right)^2 + 1}} \angle -\arctan\left(\frac{f}{f_B}\right)$$

- Thus, $H(f)$ is complex, i.e. can be written as an absolute part and an angle part

$$|H(f)| = \frac{1}{\sqrt{\left(\frac{f}{f_B}\right)^2 + 1}} \quad \text{and} \quad \angle H(f) = -\arctan\left(\frac{f}{f_B}\right)$$

Bode plot – magnitude plot

- To make a Bode-plot we need the y-axis to be in dB, thus:

- $|H(f)|_{dB} = 20 \log\left(\frac{1}{\sqrt{\left(\frac{f}{f_B}\right)^2 + 1}}\right)$

- And we need the x-axis to be logarithmic. The Bode-plot looks like this:

- **Notes:**

- When we have a pole (like f_B) the Bode plot is an asymptote that falls -20dB/decade (a decade is 10 times)
- The actual curve is 3dB below the asymptote at the cut-frequency (f_B in this example)
- (If we have a zero the Bode-plot rises with 20 dB/decade.)

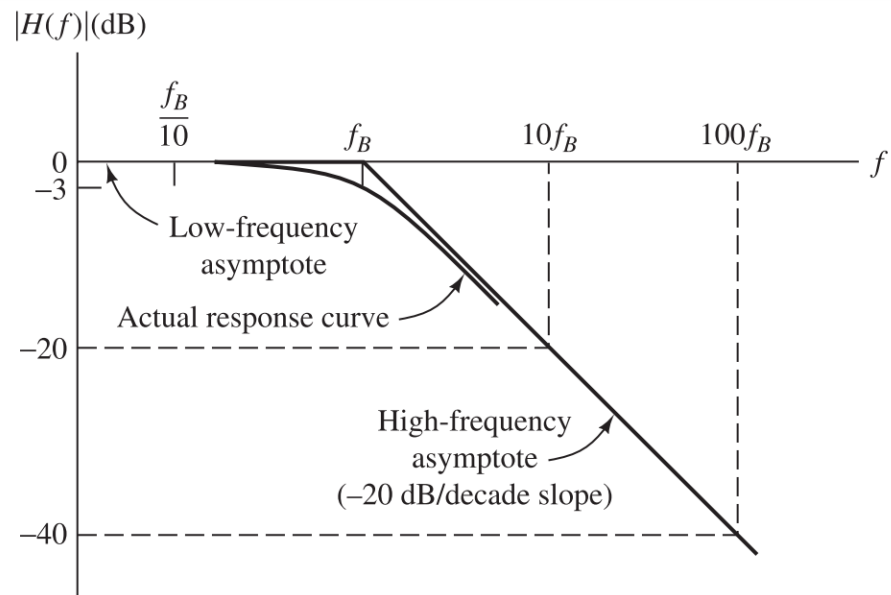


Figure 6.15 Magnitude Bode plot for the first-order lowpass filter.

Bode plot – phase plot

- However, the amplitude plot is only half the story (plot) we still miss the phase (angle) part.
- $\angle H(f) = -\arctan\left(\frac{f}{f_B}\right)$
- The x-axis must be logarithmic. The phase part of a Bode-plot looks like this:
- Notes:
 - The phase plot starts at $f_B/10$
 - The phase plot stops at $f_B \cdot 10$
 - The phase is -45° (at f_B in this example)
 - Note how the actual curve follows this approximation.

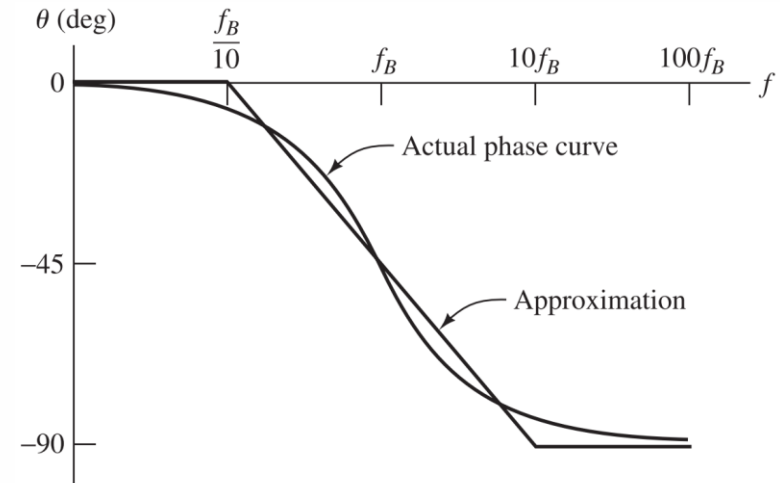


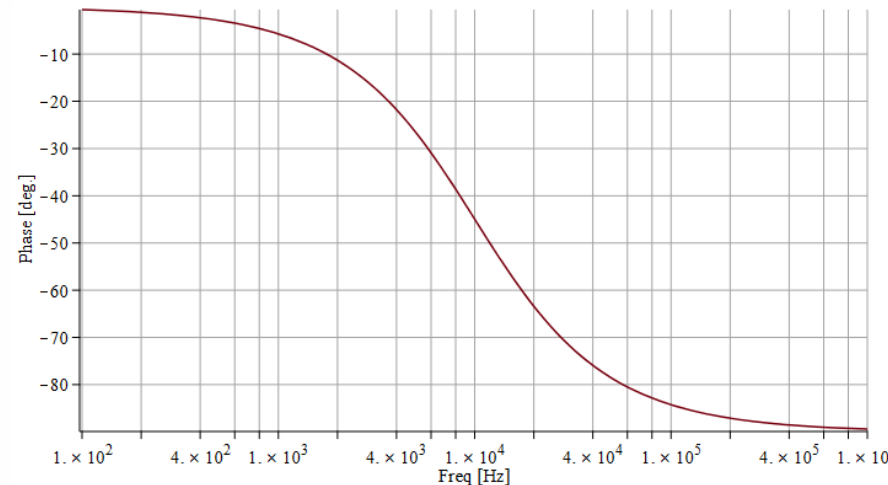
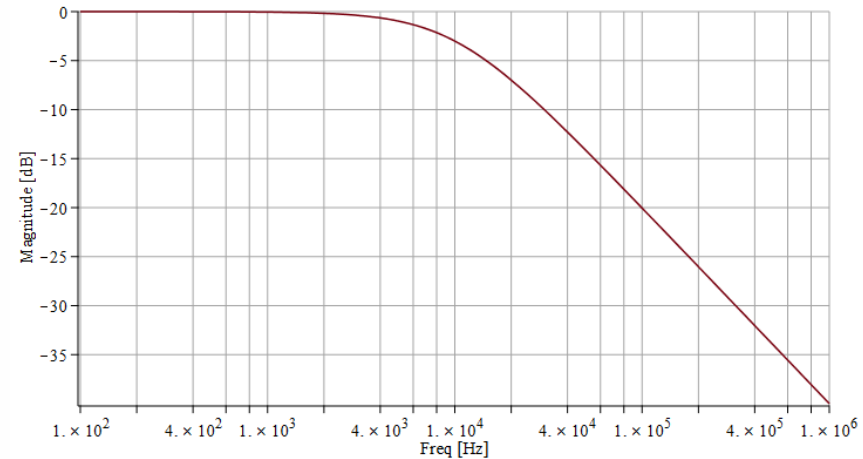
Figure 6.16 Phase Bode plot for the first-order lowpass filter.

Bode plot in Maple

- As an example assume R is 159Ω and C is 100nF . The pole is then positioned at $f_B = \frac{1}{2\pi RC} = 10000\text{Hz}$
- Maple uses this form: $H(f) = \frac{V_o}{V_i} = \frac{1}{RCs+1}$, i.e. with the Laplace operator s

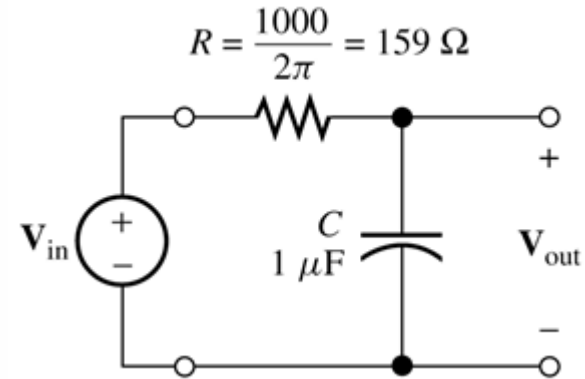
with(DynamicSystems) :

`BodePlot(TransferFunction( $\frac{1}{159 \cdot 10^{-7} \cdot s + 1}$ ), hertz = true)`

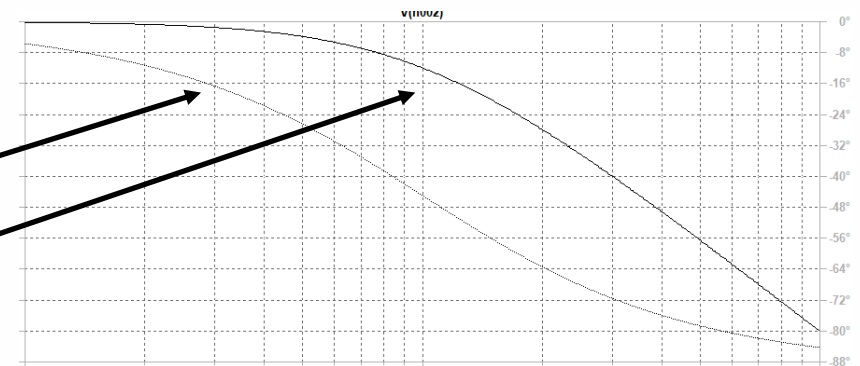


1'order lowpass filter - exercise

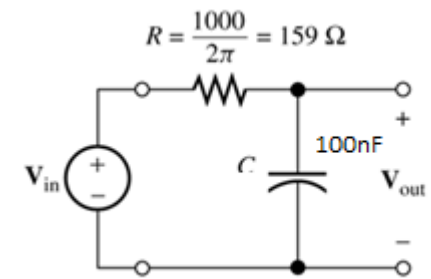
- Calculate the pole frequency (in Hertz)
- Plot $|H(f)|$ in Maple
- Draw the bode-plot (magnitude and phase asymptotes) on top of the Maple plots (use a drawing program)
 - Does the asymptotes fits with the plots ?
 - Is the actual curve 3 dB below (-3dB) the asymptote?
- Build the circuit in LT-spice and run an AC-plot (as we did with the RIAA amplifier) – does it match with the bode plots ?



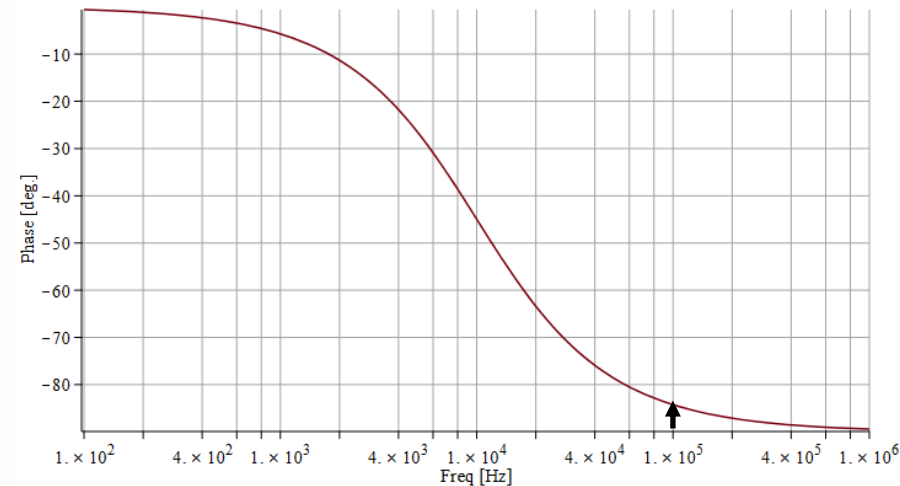
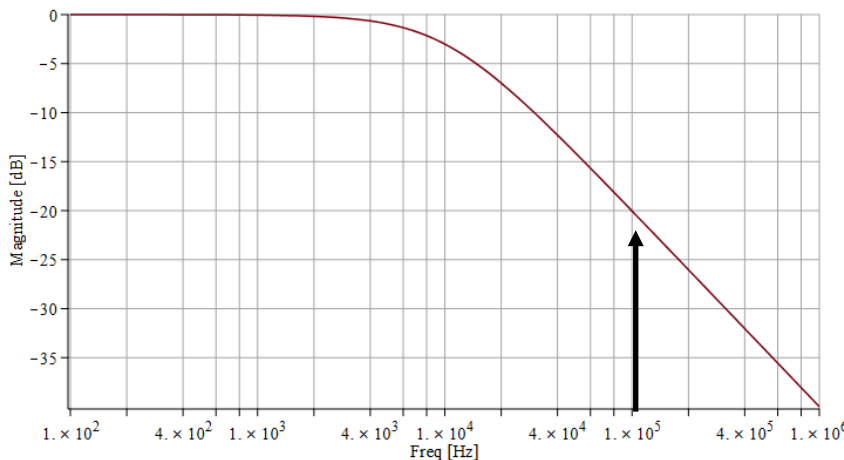
- **Note:**
 - The phase plot in LT-spice is the dashed curve
 - The amplitude plot in LT-spice is the solid curve



An example



- Assume we want a 100kHz 2 volt cosine : $V_i(t) = 2\cos(2\pi 10^5 t + 0^\circ) = 2\angle 0$ passed through the transfer-function we found earlier:



- We see that $H(f)$ attenuates the cosine with -20 dB and changes the phase of it with approx. -85° .
- When the cosine has passed $H(f)$ (actually $H(f)$ is a low-pass filter) it looks like:
- $V_i(t) = 2\angle 0$ and $H(f) = H(1000) = 0.1\angle -85^\circ$, thus $V_o(t) = 0.2\angle -85^\circ$
- i.e. $v_o(t) = 0.2\cos(2\pi 10^5 t - 85^\circ)$

Magnitude/phase plot of $H(f)$

- Applying the transfer function

- Example

$$v_{in}(t) = 5 \cos(20\pi t) + 5 \cos(200\pi t) + 5 \cos(2000\pi t)$$

$$H(10) = \frac{1}{1 + j(10/100)} = 0.9950 \angle -5.71^\circ$$

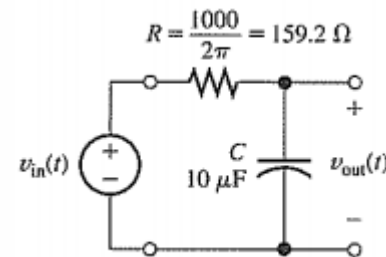
$$\begin{aligned} V_{out1} &= H(10) \times V_{in1} \\ &= (0.9950 \angle -5.71^\circ) \times (5 \angle 0^\circ) = 4.975 \angle -5.71^\circ \end{aligned}$$

$$\begin{aligned} V_{out2} &= H(100) \times V_{in2} \\ &= (0.7071 \angle -45^\circ) \times (5 \angle 0^\circ) = 3.535 \angle -45^\circ \end{aligned}$$

$$\begin{aligned} V_{out3} &= H(1000) \times V_{in3} \\ &= (0.0995 \angle -84.29^\circ) \times (5 \angle 0^\circ) = 0.4975 \angle -84.29^\circ \end{aligned}$$

$$\begin{aligned} v_{out}(t) &= 4.975 \cos(20\pi t - 5.71^\circ) + 3.535 \cos(200\pi t - 45^\circ) \\ &\quad + 0.4975 \cos(2000\pi t - 84.29^\circ) \end{aligned}$$

$$v_{in}(t) = 5 \cos(20\pi t) + 5 \cos(200\pi t) + 5 \cos(2000\pi t)$$



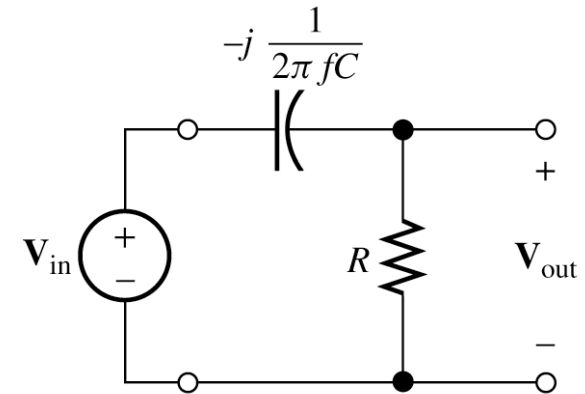
$$H(f) = \frac{V_{out}}{V_{in}} = \frac{1}{1 + j2\pi fRC}$$

$$H(f) = \frac{1}{1 + j(f/f_B)}$$

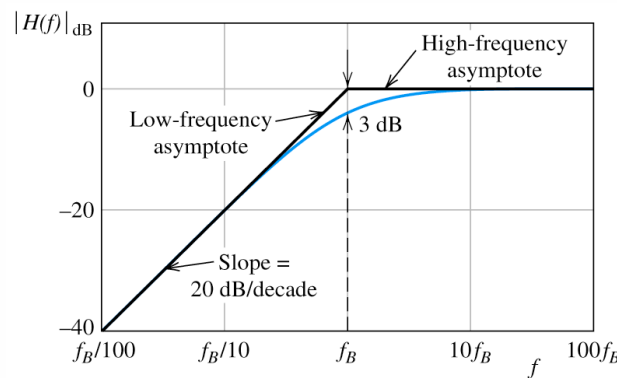
$$f_B = \frac{1}{2\pi RC} = \frac{1}{2\pi \times (1000/2\pi) \times 10 \times 10^{-6}} = 100 \text{ Hz}$$

1' order highpass filters

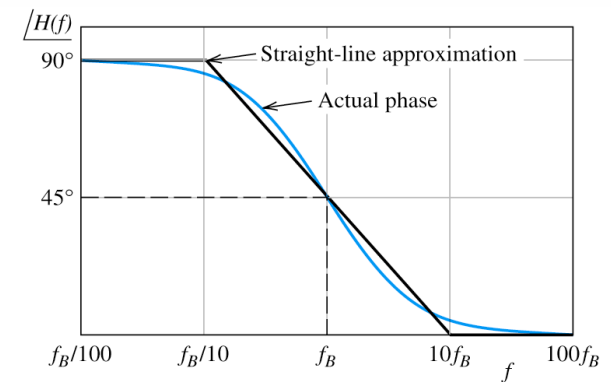
- Intuitively, remember a capacitor is open (impedance is infinite) at DC (low frequencies) and a short circuit at high frequencies. Thus, low frequencies are stopped by the capacitor, whereas high frequencies are allowed to pass –i.e. a “high-pass” filter.



$$H(f) = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{j(f/f_B)}{1 + j(f/f_B)} \quad f_B = \frac{1}{2\pi RC}$$



(a) Magnitude

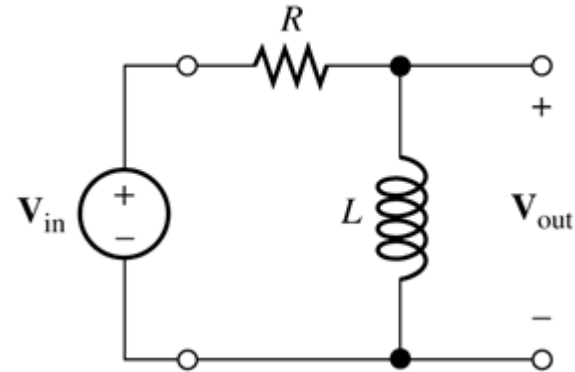


(b) Phase

Figure 6.21 Bode plots for the first-order highpass filter.

Exercise filters

- What type of filter is this and why?
- (Intuitive hint: a coil allows low frequencies to pass, but stops high frequencies)



- Find:
 - The transfer function $H(f)$
 - The pole (f_p)
 - Draw the bode plot
 - Build the circuit in LT-spice and make an AC-simulation – does the simulated results agree with the bode-plots?

Exercise

- Show the following circuits has this transfer function:

$$H(s) = \frac{1}{1 + C_1 C_2 R_1 R_2 s^2 + ((R_1 + R_2) C_2 + C_1 R_1) s}$$

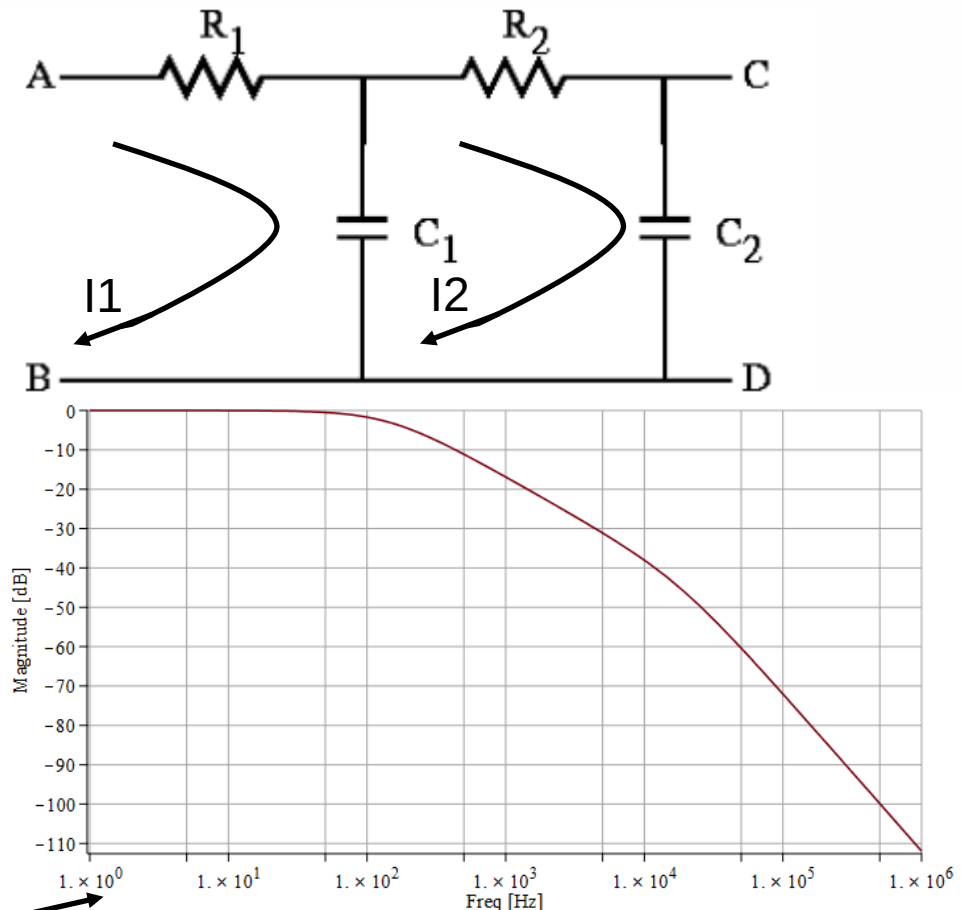
- Insert the following values:

- $C_1 = 100\text{nF}$
- $C_2 = 10\text{nF}$
- $R_1 = 10\text{k}\Omega$
- $R_2 = 1\text{k}\Omega$

- Find the roots and bring the equation on this form (Hint use "solve" in Maple to find the roots):

$$H(s) = \frac{1}{(s-p_1)(s-p_2)} \text{ where } p_x = 2\pi f_x$$

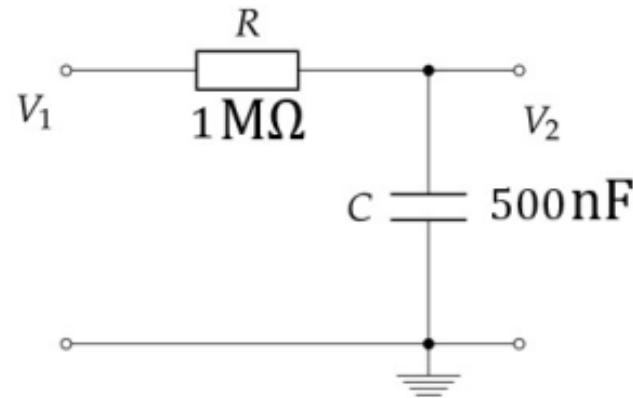
- Draw the Bode-plot (only the amplitude plot) and compare with the real curve



Note how the curve attenuate with 20 dB/decade when it passes the first pole and then increases to 40dB/decade when it passes the second pole.

Earlier Exam Exercise

Question 42



Kredsløbets overføringsfunktion $T(s) = \frac{V_2}{V_1}$ har en pol, hvilken knækfrekvens ω_0 giver denne pol anledning til?

☐ $\omega_0 = 2 \frac{\text{rad}}{\text{s}}$

☐ $\omega_0 = 1 \frac{\text{rad}}{\text{s}}$

☐ $\omega_0 = 20 \frac{\text{rad}}{\text{s}}$

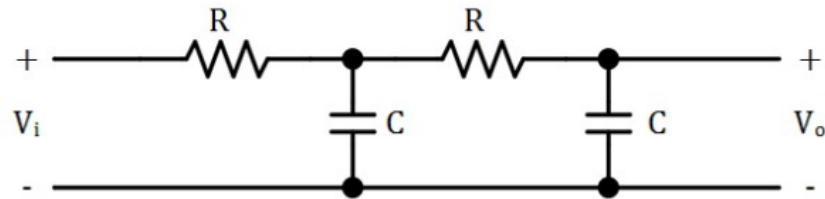
☐ $\omega_0 = 200 \frac{\text{rad}}{\text{s}}$

☐ $\omega_0 = 0.5 \frac{\text{rad}}{\text{s}}$

Earlier Exam Exercise

Question 17

RC-filter-kredsløb.



Filter-kredsløbets overføringsfunktion er defineret som:

$$H(s) = \frac{V_o}{V_i}$$

Hvilken af følgende overføringsfunktioner passer til kredsløbet?

☐ $\frac{1}{s^2 C^2 R^2 + 3RCs + 1}$

☐ $\frac{s/R}{s^2 C^2 R + 4Cs + 1/R}$

☐ $\frac{1}{(sRC + 1)^2}$

☐ $\frac{1/(RC)^2}{s^2 + 2RCs + 1/(RC)^2}$

☐ $\frac{1}{(s + RC)(s + RC)}$

The Second-Order LCR Resonator

- Let's consider a “standard” second order transfer function. This can be used to generalize and analyse many 2'nd order systems.
- We note the Q-factor which for an inductor or a tuned circuit gives an indication of its performance in a resonator circuit.
 - The Q or quality factor is a dimensionless number that describes the damping in the circuit.
 - Provides an indication of the resonans bandwidth relative to the center frequency.
 - It is used with many RF tuned circuits to indicate their performance in e.g. an oscillator or other form of resonant circuit.
- ω_0 is the resonance frequency of the circuit.
 - Very important to lookup function minimum or maximum

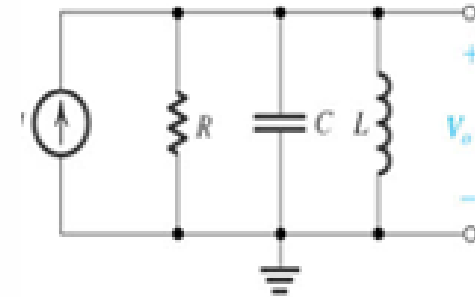
$$T(s) = \frac{a_2 s^2 + a_1 s + a_0}{s^2 + (\omega_0/Q)s + \omega_0^2}$$

The Second-Order LCR Resonator

- First, we find the transfer function for this 2nd order circuit (2nd order because it contains both a capacitor and a coil) – we use the parallel formula:

$$V_o = I \frac{1}{\frac{1}{R} + C \cdot s + \frac{1}{L \cdot s}} = \frac{I \cdot s}{\frac{s}{R} + C \cdot s^2 + \frac{1}{L}} = \frac{1}{C} \frac{I \cdot s}{s^2 + \frac{s}{R \cdot C} + \frac{1}{L \cdot C}}$$

$$\frac{V_o}{I} = \frac{1}{C} \frac{s}{s^2 + \frac{s}{R \cdot C} + \frac{1}{L \cdot C}}$$



- Now, compare this equation to the “standard” equation:

$$T(s) = \frac{a_2 s^2 + a_1 s + a_0}{s^2 + (\omega_0/Q)s + \omega_0^2}$$

Solving these equations for f_0 and Q yields:

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \quad Q_p = \frac{R}{2\pi f_0 L} \quad Q_p = 2\pi f_0 CR$$

$$\frac{V_o}{I} = \frac{1}{C} \frac{s}{s^2 + \frac{s}{R \cdot C} + \frac{1}{L \cdot C}}$$

$$\omega_0/Q = 1/CR$$

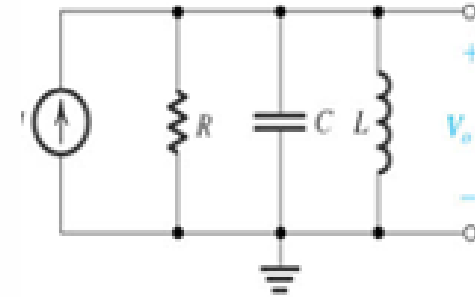
$$\omega_0^2 = 1/LC$$

- We can identify Q and ω_0

Second-Order LCR Resonator **Exercise**

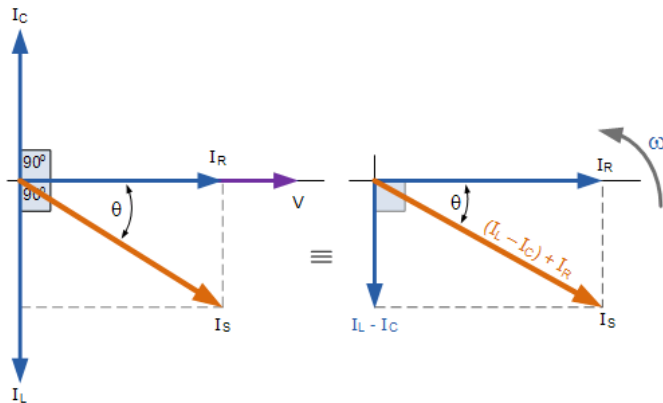
□

CX C

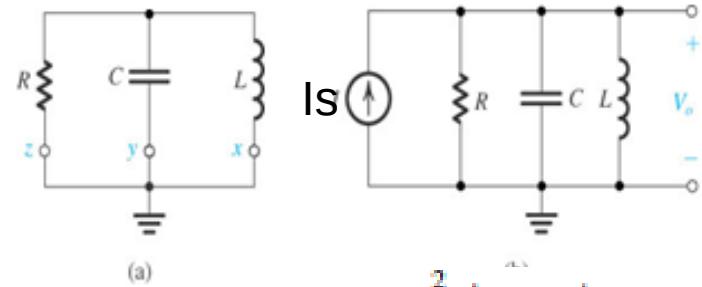


Second-Order LCR parallel resonator

- Note the magnitude and phase of the currents:
 - Low frequencies the current in the coil dominates
 - High frequencies the current in the capacitor dominates



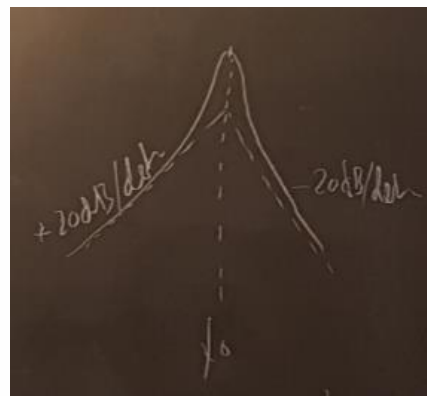
Voltage across the parallel resonant circuit for a constant variable-frequency current source



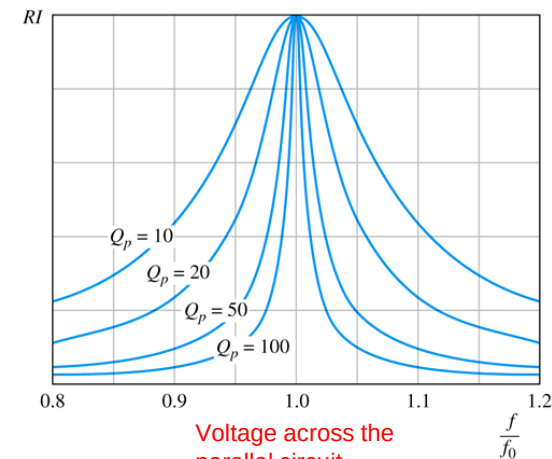
$$T(s) = \frac{a_2 s^2 + a_1 s + a_0}{s^2 + (\omega_0/Q)s + \omega_0^2}$$

$$\omega_0^2 = 1/LC \quad \omega_0/Q = 1/CR$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \quad Q_p = \frac{R}{2\pi f_0 L} \quad Q_p = 2\pi f_0 CR$$



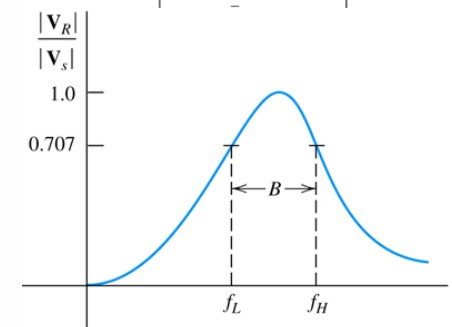
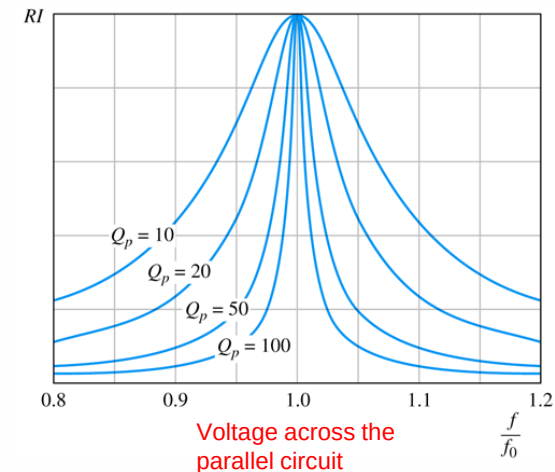
Bode plot of T(s)



Voltage across the parallel circuit

Using the parallel resonator as BP-filter

- Resonators can be used as bandpass filters
- The bandwidth (B) is the difference between the half power frequencies

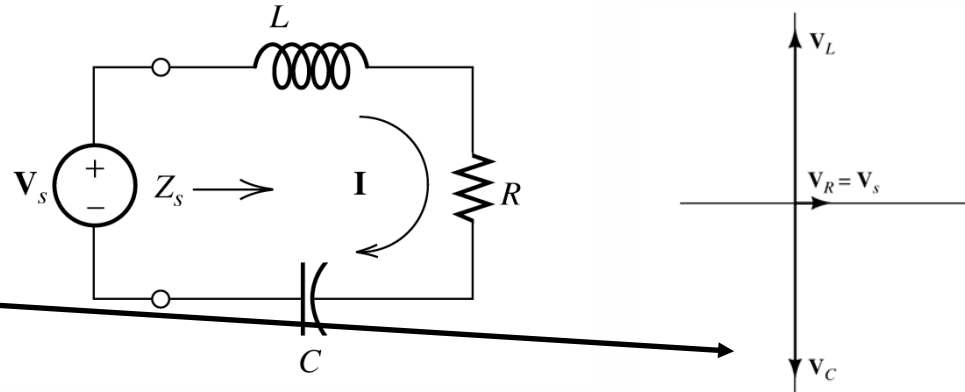


$$B = f_H - f_L \quad f_H \cong f_0 + \frac{B}{2}$$

$$B = \frac{f_0}{Q} \quad f_L \cong f_0 - \frac{B}{2}$$

Second-Order LCR serial resonator

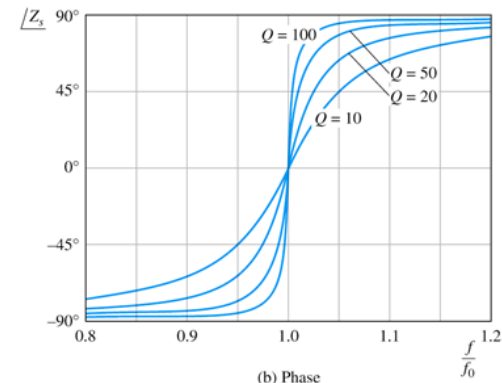
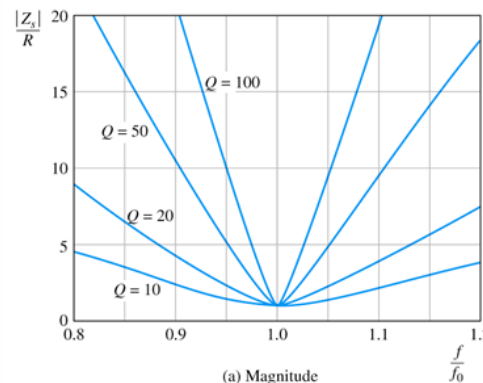
- The serial resonance circuit
- Note the voltages over the components and their phase



$$f_0 = \frac{1}{2\pi\sqrt{LC}} \quad Q_s = \frac{1}{2\pi f_0 CR} \quad Q_s = \frac{2\pi f_0 L}{R}$$

These Q's are for a serial resistor only!

- Normalized magnitude and phase of the impedance



Realization of lowpass filter

- The second order transfer function can be written by using the voltage divider rule:

$$T(s) \equiv \frac{V_o}{V_i} = \frac{Z_2}{Z_1 + Z_2} = \frac{Y_1}{Y_1 + Y_2} = \frac{1/sL}{(1/sL) + sC + (1/R)}$$

$$= \frac{1/LC}{s^2 + s(1/CR) + (1/LC)}$$

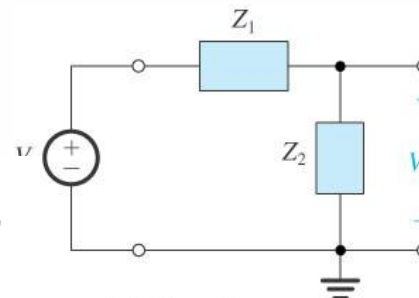
$$T(s) = \frac{a_2 s^2 + a_1 s + a_0}{s^2 + (\omega_0/Q)s + \omega_0^2}$$

$$\omega_0^2 = 1/LC \quad \omega_0/Q = 1/CR$$

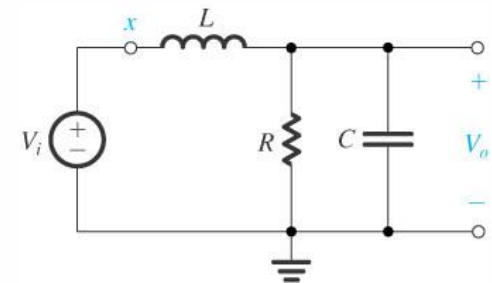
Or:

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \quad Q_p = \frac{R}{2\pi f_0 L} \quad Q_p = 2\pi f_0 CR$$

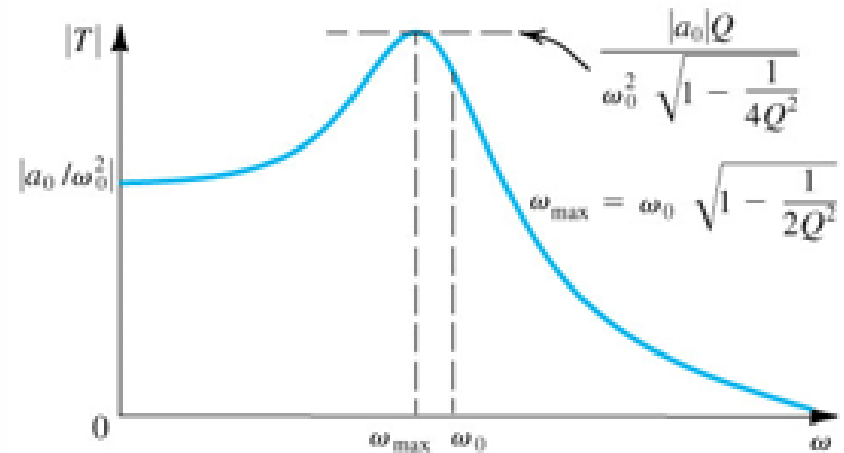
$$T(s) = \frac{V_o(s)}{V_i(s)} = \frac{Z_2(s)}{Z_1(s) + Z_2(s)}$$



(a) General structure



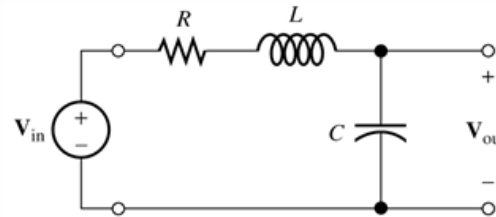
(b) LP



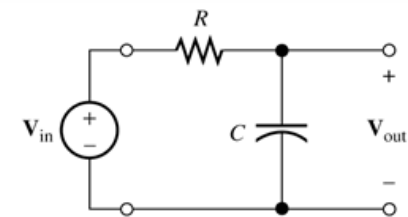
Try: $Q=100$ – how large is the peak compared to the value at $\omega=0$

Realization of lowpass filter

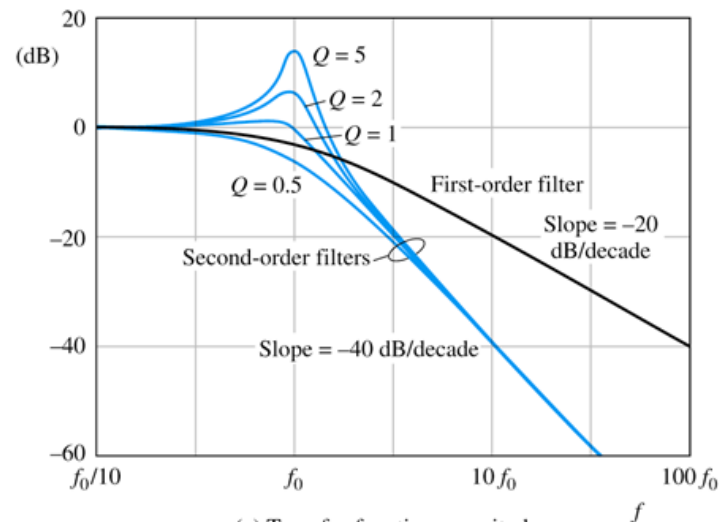
- **EXERCISE**
- Find the transfer function $H(f)$ and ω_0 for both circuits.
- Find Q in (a).
- For both circuits draw bode-plot (only magnitude - (b) only for $Q = 0.707$).



(a) Second-order lowpass filter



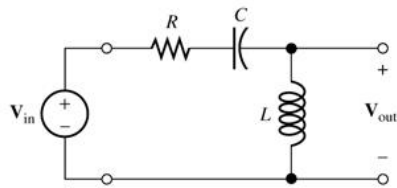
(b) First-order lowpass filter



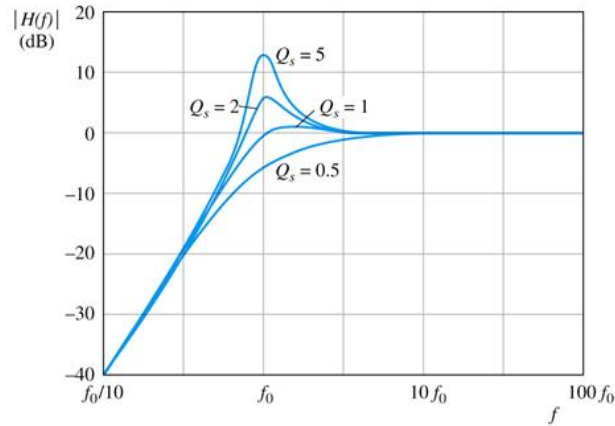
(c) Transfer-function magnitudes

Realization of high-pass filter

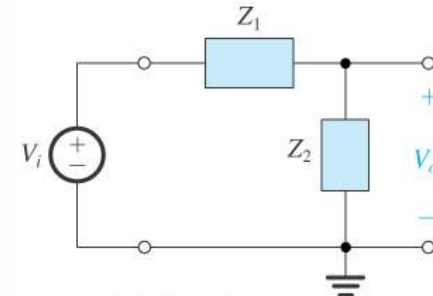
□



(a) Circuit diagram



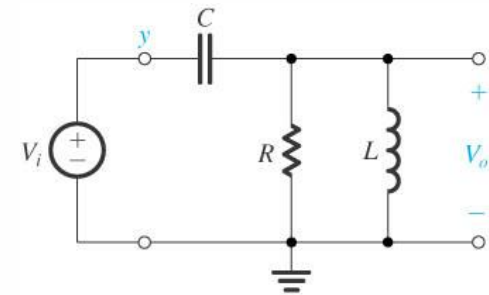
(b) Transfer-function magnitude



(a) General structure

$$T(S) = \frac{Z_2}{Z_2 + Z_1} = \frac{Y_1}{Y_1 + Y_2} = \frac{Cs}{Cs + \frac{1}{Ls} + \frac{1}{R}} = \frac{RLCs^2}{RLCs^2 + R + Ls} = \frac{s^2}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}$$

$$T(s) \equiv \frac{V_o}{V_i} = \frac{a_2 s^2}{s^2 + s(\omega_0/Q) + \omega_0^2}$$



(c) HP